

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1777

Palm-Hackberry 138 kV
June 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1777. The Interconnection Customer proposes a 73.4 MW solar plus storage facility interconnecting at the Palm-Hackberry 138 kV substation in the Pennsylvania area. The request was received on June 09, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 619 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$800,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Firstenergy Springdale Power Sta Substation-Gobain 138 KV
Contingency: Allegheny Energy Units 3 4 & 5 345-1 kV TX
Loading: 117.4% (Rating: 192 MVA)
DFAX: 11.7%
Pre-project loading: 112.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Firstenergy Springdale Power Sta Substation-Gobain 138 KV	Rebuild line with higher-capacity conductor	\$1,536,000	52% (\$800,000)
	Total allocated to project		\$800,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.